

SERVICE RUNBOOK

Search Service

service: search-service

TIER 2

ON-CALL

1 Service Overview

`search-service` powers product and content search, backed by an **indexing pipeline** that consumes catalog update events and a **query-caching layer** for performance.

IMPACT Degraded search directly affects product discovery and conversion. Stale results that touch **pricing accuracy** carry customer-facing financial risk.

2 Common Failure Patterns

Stale search results are most often a cache invalidation issue — check whether recent catalog updates arrived in unusually large batches, since batch handling has historically had edge cases around invalidation triggers.

Latency regressions are frequently caused by new query paths introduced by feature work that aren't covered by existing database indexes, leading to full collection scans under load.

Indexing lag (new items not appearing in search) has been linked to consumer group rebalance loops in the upstream event pipeline resetting offsets repeatedly.

SYMPTOM	LIKELY CAUSE / FIRST CHECK
Stale / outdated results after catalog update	Cache invalidation skipped — check batch size handling
Elevated response latency, no deploy correlation	Missing index for a new query path — check recent feature launches

SYMPTOM	LIKELY CAUSE / FIRST CHECK
New items missing from search	Indexing pipeline lag — check consumer group rebalance status
Gradual memory growth post-deploy	Query cache not respecting configured max size

■ CRITICAL / FINANCIAL IMPACT ■ DEGRADATION

3 Diagnostic Steps

- 1

Check whether a recent **catalog update batch** exceeded typical size thresholds.
- 2

Review **query plans** for recently added filters or query paths for missing indexes.
- 3

Check **indexing pipeline consumer lag** metrics if new content is not appearing.

4 Escalation Criteria

- ESCALATE IF:
- Search latency exceeds **3× baseline for more than 15 minutes**.
 - Stale-result issues **affect pricing accuracy** (customer-facing financial impact).